Christopher Martin

Part 1

n	$A(n) = (1 + 1/n)^n$
1	2
2	2.25
4	2.44140625
12	2.61303529
100	2.704813829
1,000	2.716923932
10,000	2.718145927

Step 2: $n \to \infty$
$A(n) = 2.71828$

Part 2

k	$k!$
0	1
1	$1 \times 1 = 1$
2	$2 \times 1 = 2$
3	$3 \times 2 \times 1 = 6$
4	$4 \times 3 \times 2 \times 1 = 24$
5	$5 \times 4 \times 3 \times 2 \times 1 = 120$
6	$6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$
7	$7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 5{,}040$

-Step 4:

k	Term $(1/k!)$	Running total
0	$\frac{1}{0!} = \frac{1}{1} = 1$	1.00
1	$\frac{1}{1!} = \frac{1}{1} = 1$	2.00
2	$\frac{1}{2!} = \frac{1}{2} = 0.5$	2.50
3	$\frac{1}{3!} = \frac{1}{6} \approx 0.16667$	2.66667
4	$\frac{1}{4!} = \frac{1}{24} \approx 0.04167$	2.70834
5	$\frac{1}{5!} = \frac{1}{120} \approx 0.00833$	2.71667
6	$\frac{1}{6!} = \frac{1}{720} \approx 0.00139$	2.71806
7	$\frac{1}{7!} = \frac{1}{5040} \approx 0.00020$	2.71826

8	$\frac{1}{8!} = \frac{1}{40{,}320} \approx 0.00002$	2.71828
9	$\frac{1}{9!} = \frac{1}{362{,}880} \approx 0.000002$	2.718282

Q4a: Took 9 terms for the running total to completely stop changing at the 5th decimal place. Factorials grow much faster than the powers of n from Part 1

Part 3

- Step 5: $|2.718145927 - 2.71828| = 0.00013407$

The factorial series yields a closer approximation to the value of e than the compound interest formula w/ 10,000 iterations

- Step 6: Business analytics models rely on continuous growth because real world variables like consumer behaviors, market changes, and asset values update constantly rather than in rigid, discrete intervals. Modeling inputs in e allows analysts to create smoother, more mathematically precise forecasts for predicting long term trends & managing financial risk. For instance, you'll encounter this constant in continuous growth & decay models, which are essential for tracking things like customer churn or calculating depreciation of corporate assets over time.